

Title: Student Report Automation System using UiPath

1. Introduction

This project focuses on automating the process of generating and sending student academic reports via email using UiPath. The system reads student data from an Excel sheet, calculates/uses marks and attendance, and sends personalized performance reports to each student through SMTP email.

2. Objective

- To automate student report generation
- To reduce manual emailing work
- To ensure fast and error-free communication of academic performance
- To improve efficiency using RPA (Robotic Process Automation)

3. Tools & Technologies Used

- UiPath Studio
- Microsoft Excel
- SMTP Email (Gmail)
- VB.NET expressions (inside UiPath)

4. Input Data (Excel Structure)

The Excel sheet contains the following columns:

- Name
- Email
- Maths
- Science
- English
- Attendance
- Total

Each row represents a student's academic record.

Name	Email	Maths	Science	English	Attendance	Total
Mrunal	mrunalarote05@gmail.com	80	75	70	90	225
Jyoti	jotiarote@gmail.com	60	65	58	85	183
Amit Verma	prasadarote00@gmail.com	45	50	40	70	135
Priya Singh	priya@gmail.com	90	88	92	95	270
Rohit Kumar	rohit@gmail.com	72	68	70	88	210
Neha Joshi	neha@gmail.com	55	60	58	80	173
Karan Mehta	karan@gmail.com	35	40	38	65	113
Anjali Desai	anjali@gmail.com	85	82	80	92	247
Vikas Yadav	vikas@gmail.com	67	70	65	85	202
Pooja Nair	pooja@gmail.com	78	75	80	89	233

5. Workflow Description

Step 1: Read Excel Data

- The Excel file is read using **Read Range** activity
- Data is stored in a DataTable (dt)

Use Excel File

Workbook path *
 {} "C:\uipathproject\students.xlsx"

Reference as *
 Excel

Create if not exists: ☒ On
 Save changes: ☒ On

Do

Read Range

Range *
 {} Excel.Sheet("sheet1")

Has headers: ☒ On
 Visible rows only: ☒ On

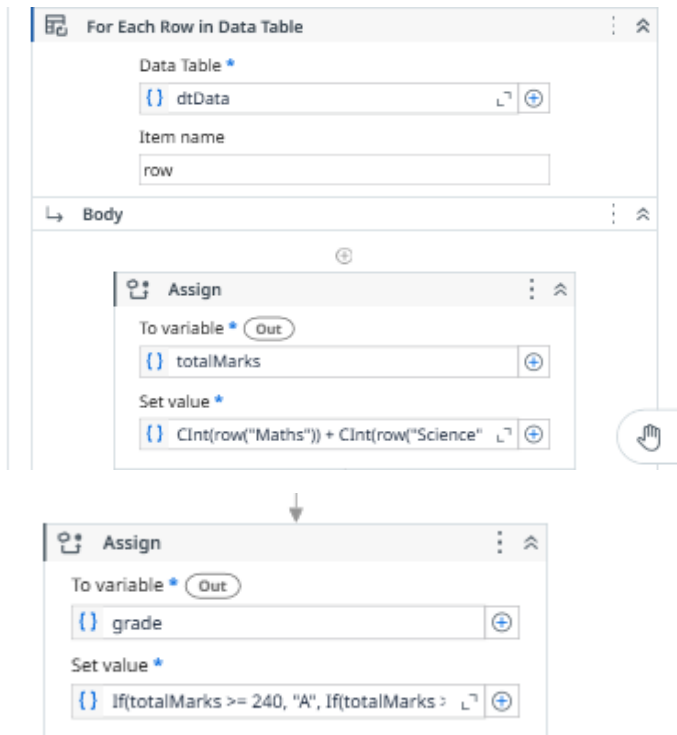
Data Manager | Output | Markers | Terminal | Errors

(x) Variables | (a) Arguments | Namespaces | Connections

Name	Data Type	Scope	Default Value
Create variable			
{x} dtData	DataTable	Main	{}

Step 2: Loop Through Each Student

- A **For Each Row** loop is used
- Each student record is processed individually

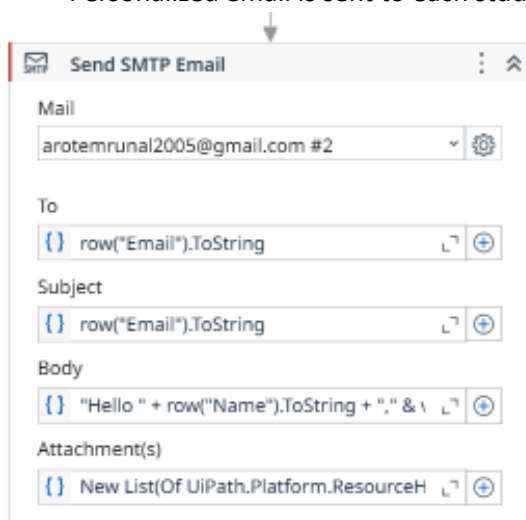


Step 3: Generate Report Content

- Student details and marks are extracted
- Total marks are displayed (or calculated if needed)

Step 4: Send Email via SMTP

- Gmail SMTP configuration is used
- Personalized email is sent to each student using their email ID



Connect to Mail



Connecting allows UiPath products to interact with Mail data on your behalf. This may include read, write, modify and delete data, depending on your Mail permissions.

[Ask Autopilot for help](#)

Folder in which connection is stored ⓘ

My Workspace

Authentication Type

SMTP & POP3 Username and Password

Server Address *

smtp.gmail.com

Port *

587

Email *

arotemrunal2005@gmail.com

Password *

.....

Ignore Certificate Revocation List *

☐ True

☒ False

Secure connection *

Start TLS

Connect

The current session will timeout in 11:16

[Terms](#)

[Privacy](#)

[Help](#)

6. Email Format Example

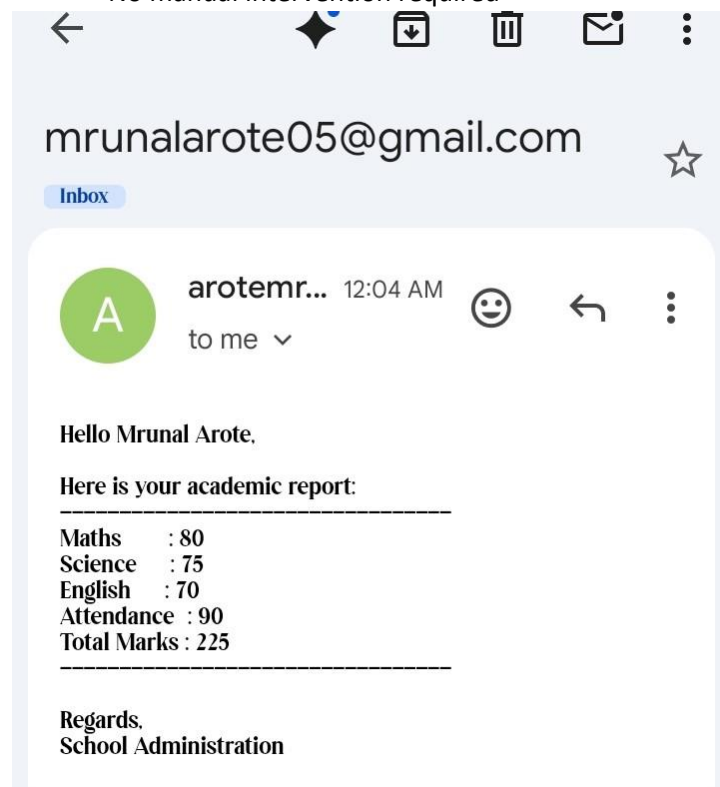
Each student receives an email like:

- Greeting with Name
- Subject-wise marks
- Attendance details
- Total marks
- Closing message

```
1 |"Hello " + row("Name").ToString + "," & vbCrLf &  
2 vbCrLf &  
3 "Here is your academic report:" & vbCrLf &  
4 "-----" & vbCrLf &  
5 "Maths    : " + row("Maths").ToString & vbCrLf &  
6 "Science  : " + row("Science").ToString & vbCrLf &  
7 "English   : " + row("English").ToString & vbCrLf &  
8 "Attendance : " + row("Attendance").ToString & vbCrLf &  
9 "Total Marks : " + row("Total").ToString & vbCrLf &  
10 "-----" & vbCrLf &  
11 vbCrLf &  
12 "Regards," & vbCrLf &  
13 "School Administration"
```

7. Output

- Each student receives an individual email report
- Data is automatically fetched from Excel
- No manual intervention required





8. Advantages

- Saves time and effort
- Reduces human errors
- Fast communication of results
- Scalable for large number of students
- Easy to maintain and modify

9. Conclusion

The Student Report Automation System successfully demonstrates how UiPath can be used to automate repetitive academic tasks. It improves efficiency by automatically reading student data from Excel and sending personalized reports via email. This project highlights the practical use of Robotic Process Automation in educational environments.

10. Future Enhancements

- Add Grade calculation (A/B/C/D)
- Generate PDF report cards
- Add dashboard for performance analysis
- Integrate database instead of Excel
- Add email attachments